

Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; Cpk, one-sided process capability index; CQA, critical quality attribute; CV, column volume; DNA, deoxyribonucleic acid; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; Fc, fragment crystallizable region of the antibody; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; IgG1, an immunoglobulin G subclass; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSM, response surface methodology; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

Executive summary

Protein A affinity chromatography is the capture step of the A-Mab drug substance process. This report presents the characterization study executed on the qualified scale-down model under the plan PCP-005, in support of transfer of the process to commercial manufacture at a production bioreactor scale of 15,000 L. The scope of the study was set by the risk assessment RA-001. 6 process parameters were characterized. 4 were studied in a multivariate design and 2 were assessed one at a time.

The screening design was a two-level full factorial in 4 parameters with 3 centre points, 19 runs in all. The response-surface design was a face-centred central composite in the same parameters, 28 runs. Three responses were measured on every eluate pool, pool host cell protein, step yield and leached Protein A.

Pool host cell protein is governed by protein load and by elution buffer pH. It rises with load, it falls as elution pH rises, and the interaction between the two is significant ($p < 0.001$). The response-surface model for pool host cell protein has an R^2 of 0.989 and shows no significant lack of fit. Step yield is raised by extending the end of pool collection and is lowered by protein load and by load flow rate. Leached Protein A showed no significant effect of any parameter across the ranges studied, which is what the leaching mechanism predicts within a single run.

The design space is the part of the characterized region in which pool host cell protein stays at or below the step's in-process limit of 10,724 ng/mg and leached Protein A stays at or below 5 ppm. On an evenly spaced grid across the four characterized ranges, 73.4% of the region meets both criteria, and pool host cell protein is the binding response. The excluded part is the corner at high protein load and low elution pH. The same corner is the only part of the normal operating ranges that breaches the limit, and at the low edge of the elution pH range the mean model reaches the limit at a load of 33.9 g/L resin, below the upper edge of the load range of 40 g/L resin. Protein load and elution buffer pH are therefore taken forward as a jointly controlled pair (§6).

Protein load and elution buffer pH were classified as well-controlled critical process parameters. Load flow rate and end of pool collect were classified as key process parameters, and operating temperature and bed height as general process parameters. At commercial scale the drug substance meets its criteria for the three impurity attributes this step governs, and the tightest of the three capabilities is host cell protein at a Cpk of 6.14. Two deviations were recorded during execution and both were retained after investigation (§13). This report rolls up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in fed-batch mammalian cell culture and purified by the Novacyte Biologics platform process (CMC Biotech Working Group 2009). Its principal mechanism of action is antibody-dependent cellular cytotoxicity, so the glycan attributes formed in the production bioreactor are linked to efficacy and are characterized in PCR-003. The purification train is Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration, and ultrafiltration with diafiltration. Protein A chromatography is Step 5 of the drug substance process and the first chromatographic operation in it.

The step is operated in bind and elute mode on an immobilized staphylococcal Protein A resin. The ligand binds the Fc region of the antibody at neutral pH through hydrophobic and hydrogen bonding contacts at the interface between the second and third constant domains of the heavy chain. Host cell protein, DNA and most other components of the clarified harvest pass into the flow-through or are removed in the wash. Lowering the pH protonates histidine residues at the interface between the Fc region and the ligand, weakens the interaction and releases the antibody into a small volume of low-pH eluate. The step therefore concentrates the product, removes the bulk of the process impurities, and introduces one impurity of its own, the ligand that leaches from the resin.

The step sets one critical quality attribute, leached Protein A, and it delivers the principal removal of host cell protein and of DNA in the train. The eluate is the feed to the low-pH viral inactivation step, which is characterized in PCR-006. No viral clearance claim is made for Protein A chromatography in this package. The cumulative viral clearance claim rests on the low-pH hold, on anion exchange and on small-virus retentive filtration, and it is consolidated in PCMR-001.

The commercial drug substance facility is Novacyte Biologics, Grafton, WI. The characterization studies reported here were executed at Novacyte Biologics, Cambridge, MA, on the scale-down model described in §3.1. The related documents of the package are listed in Table 1.1.

Table 1.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope

Docu- ment ID	Title	Relationship
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-005	Process Characterization Plan (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

1.2 Objectives

The study had five objectives.

1. To quantify the relationship between each characterized process parameter and the quality and performance attributes the capture step governs.
2. To define the operating region over which those attributes meet the criteria that apply to them.
3. To establish a proven acceptable range for each parameter and to place the normal operating range inside it.
4. To classify each parameter under SOP-4001 from the demonstrated effects and the control capability of the commercial equipment.
5. To justify the ranges taken forward to Stage 2 and the contribution this step makes to the control strategy.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8, in which a design space is established from the demonstrated relationships between parameters and quality attributes, and movement within that space is not a change (International Council for Harmonisation 2009). Criticality is treated as a continuum and not as a binary classification, following ICH Q9 and the A-Mab case study, and the attributes at high and very high criticality are called critical (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). The link from process understanding to the control strategy follows ICH Q11 (International Council for Harmonisation 2012).

ICH Q5A governs the viral safety evaluation of the process (International Council for Harmonisation 2023a). It is not applied in this report, because no viral clearance is claimed for this step. The controlled documents that govern execution and analysis are listed in Table 1.2.

Table 1.2: Controlled documents and validated methods for the capture step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The Novacyte downstream platform has produced three prior humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab, and the Protein A capture step has been operated at commercial scale for all three. The resin, the buffer system and the operating principle are unchanged for A-Mab. The platform history therefore establishes the mechanism of the step and the approximate size of the parameter effects, and the purpose of this study is to bound them for A-Mab rather than to discover them. Some acceptance limits used in this package come from X-Mab clinical experience, and they are recorded in the drug substance quality attribute register rather than in this report.

Platform knowledge was not treated as transferable in one respect. An alternate Protein A resin was assessed under RA-004, which concluded that independent characterization would be required for it and that bridging from platform data would not be accepted. This study covers the platform resin only, and its conclusions do not extend to the alternate resin.

The physical chemistry of the step sets the expectation against which the results were read. Protein load is the mass of antibody applied per volume of resin, and as it approaches the dynamic binding capacity the mass transfer zone reaches the column outlet, antibody breaks through into the flow-through, and impurity that the wash would otherwise displace is carried into the bed. Elution pH sets how completely and how quickly the interaction between the Fc region and the ligand is disrupted, so a lower pH releases the antibody sooner and in a sharper peak, and it releases with it the host cell protein held weakly on the resin or on the antibody. Load and elution pH therefore act on the same quantity, the impurity burden carried into the eluate, and an interaction between them was expected before the study was run. Load flow rate sets the residence time and hence how far into the pore of the bead the antibody diffuses before the fluid moves on, so a faster load broadens the mass transfer zone and costs yield through earlier breakthrough. Extending the end of pool collection recovers antibody from the tail of the elution peak and dilutes the pool.

Leached Protein A was expected to be flat across the operating ranges. Ligand leaves the resin by hydrolysis of the immobilization chemistry and by proteolysis of exposed ligand, and the leached fragments elute with the antibody because they still bind the Fc region. The amount depends on the coupling chemistry of the resin, on the cumulative cycle number and on the sanitization history, and all three are properties of the resin over its life rather than of a single run.

2.2 Quality attributes in scope

The capture step governs three process impurity attributes of the drug substance. They are listed with their criticality in Table 2.1. Leached Protein A is introduced by this step and is not present before it. Host cell protein and DNA enter with the clarified harvest and are removed here, and both are removed further by cation exchange (PCR-007) and anion exchange (PCR-008).

Table 2.1: Quality attributes governed by the capture step, with the drug substance acceptance criteria and the criticality assigned in RA-001.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0-5 ppm	L-M	16
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

The Tool #1 score in Table 2.1 is the product of impact and uncertainty from the criticality assessment of the A-Mab case study, and the criticality level is the continuum position derived from it (CMC Biotech Working Group 2009). Host cell protein carries the highest score of the three and is moderate to high in criticality, because residual host cell protein is immunogenic and its level in the drug substance depends on the performance of three chromatographic steps in series. Leached Protein A is low to moderate, because the ligand is cleared efficiently downstream and the observed levels in the drug substance are far below the criterion. Residual DNA is very low in criticality, because the polyanionic molecule does not bind the ligand and is removed in the flow-through and the wash.

Two of these attributes were measured as responses in the designed experiments, pool host cell protein and leached Protein A. DNA in the pool was not a designed response. Residual DNA is measured by qPCR under AMV-3014 as part of the pool testing, and no log reduction is claimed for this step in this report. The drug substance capability for residual DNA is given in §8.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked the process parameters of the capture step by their potential impact on the quality attributes in Table 2.1 and by their potential to interact with other parameters, and it filtered them into three groups. Parameters with a high combined rank were taken into the multivariate designs. Parameters with a moderate rank, whose ranges could be supported one at a time, were taken into univariate assessment. Parameters controlled to a platform specification and carrying no plausible route to a quality attribute were not studied, and their ranges are held at the platform values.

4 parameters entered the multivariate design, protein load, elution buffer pH, load flow rate and end of pool collect. Protein load and elution buffer pH were ranked highest because both act directly on the impurity burden of the eluate. Load flow rate and end of pool collect were included because both act on yield and both can interact with load through the mass transfer zone. 2 parameters, operating temperature and bed height, were assessed one at a time, and the reason for each is given in §4.4.

The characterization ranges were set wider than the routine control ranges, so that the study establishes a knowledge space rather than only the region the process will occupy. The ranges are given in Table [4.1](#).

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a laboratory chromatography system qualified as a model of the commercial step under SOP-1001. The model was designed on the established principles of chromatography scaling. Bed height, linear velocity, protein load per volume of resin, and the equilibration, wash and elution volumes expressed in column volumes were held equal to the commercial values, and column efficiency was verified by plate count and peak asymmetry before each campaign. Because those quantities are held equal, the residence time and the number of transfer units of the small column match the commercial column, and the chemistry the antibody experiences is the same at both scales.

Qualification compared the model against at-scale performance for the input and output attributes of the step, which are step yield, pool host cell protein, leached Protein A and residual DNA. The comparison and its acceptance criteria are recorded in the qualification report cited by PCP-005 and are not reproduced here. The model is used in this report to support conclusions about the commercial step for those attributes, and for no others.

The centre-point replicates of each design were executed under the same conditions as the commercial set-point, and their spread is the pure error term used to test lack of fit in §5.3. That spread carries the analytical variability of the assays as well as the run to run variability of the model, so it is an upper bound on the reproducibility of the step itself.

3.2 Operation

Clarified harvest from the depth filtration train (PCR-004) was loaded onto the equilibrated column at the protein load and the linear velocity set by the design. The column was washed to remove unbound and weakly retained material, and the antibody was eluted with a low-pH buffer at the pH set by the design. Pool collection started on the rising edge of the elution peak at the platform threshold and ended at the volume set by the end of pool collect parameter, expressed in column volumes. The pool was neutralized within the platform hold time and sampled before transfer to the low-pH viral inactivation step.

Resin lot, buffer composition and the sanitization sequence were held at their platform values throughout, and the resin was operated within the cycle limits of SOP-2008.

These inputs are controlled by identity and quality testing on receipt and by the life-cycle controls of SOP-2008, and they were not varied in this study. Operation of the step follows SOP-2007.

3.3 Analytical methods

The validated methods applied at this step are listed in Table 1.2 and summarized in Appendix D. Pool host cell protein was measured by the generic process ELISA under AMV-3012, whose precision is 9.5 % and whose limit of quantitation is 1 ng/mg. Leached Protein A was measured by ELISA under AMV-3016, whose precision is 6.5 %, whose limit of quantitation is 0.25 ppm and whose accuracy is 97.4 %. Residual DNA was measured by qPCR under AMV-3014 and size variants by SEC-HPLC under AMV-3011. Step yield was calculated from the protein concentration of the load and of the pool and the corresponding volumes.

The two impurity assays differ in how much of the observed variation they contribute. The host cell protein ELISA reports a mass ratio against a total protein measurement, and its precision is the larger of the two. The leached Protein A ELISA measures a low-abundance analyte in the presence of a large excess of the antibody it binds, and 41 % of the variance observed in the reported result is attributed to the method. That figure is used in §5.1 and §13.2 when the reproducibility of the leached Protein A response is discussed.

3.4 Sampling plan

Each run was sampled at the load, at the flow-through, at the end of the wash and at the eluate pool. The eluate pool sample carries the three responses of the design and is the sample the analysis rests on. Load samples were retained so that a pool result outside its expected trend could be traced to its feed, and the flow-through and wash samples were retained for investigation and were not routinely assayed.

The screening design carried 3 centre-point replicates and the response-surface design carried 4. Replicates were distributed through the randomized run order rather than executed consecutively, so that the pure error term carries any drift across the campaign.

3.5 Statistical methods

Factors were coded so that the set-point is 0 and the edges of the characterization range are the coded levels $-1/+1$. For the four multivariate parameters the set-point coincides with the midpoint of the characterization range, so the coded centre of both designs is the commercial set-point. The screening data were fitted by ordinary least squares with all main effects and all two-factor interactions, and an effect is reported

as twice the coded coefficient, which is the change in the response across the full range of the factor. The response-surface data were fitted with the full quadratic model in the same four factors. Terms were tested at $\alpha = 0.05$.

Model adequacy was assessed by R^2 , by adjusted R^2 , by predicted R^2 from the PRESS statistic, by the overall F test, and by the lack of fit test that compares the residual variation not explained by the model with the pure error of the centre-point replicates. A response with no term significant at $\alpha = 0.05$ is reported as robust to the parameters across the ranges studied, and no model is used for it.

Two acceptance criteria apply to the responses of this step, and they have different origins. Leached Protein A is judged against the drug substance criterion of 5 ppm, because the level in the capture pool is already close to it and a backward calculation through the downstream clearance would give a criterion far wider than the pool ever reaches. Pool host cell protein is judged against an in-process limit, because the drug substance criterion of 100 ng/mg applies to material that has passed cation exchange and anion exchange and does not apply to the capture pool. The in-process limit was calculated by carrying the drug substance criterion back through the host cell protein clearance the downstream steps deliver in the nominal train, a factor of 483, and dividing the result by an assurance margin of 4.5. The limit is 10,724 ng/mg. The margin is what makes it a control rather than a break-even point, because a limit set at the undivided ceiling would leave the drug substance dependent on every downstream step delivering its nominal clearance exactly. Step yield is a process performance attribute with no acceptance criterion, and it is reported without one.

Proven acceptable ranges were computed from the fitted response-surface models in two forms. The first holds the other parameters at their set-points and reports the range of the parameter over which the mean prediction meets the criterion. The second lets the other parameters vary within their normal operating ranges by Monte-Carlo, adds the residual variability of the model, and reports the range over which the 95 % predictive interval stays within the criterion. The second is a robustness criterion and is the narrower of the two. It uses 2,000 draws at each of 81 grid points across the characterization range, under a fixed seed.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter of the train varying within its normal operating range, and is reported as a one-sided Cpk against the drug substance criterion.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the parameters, their set-points, their normal operating ranges, the ranges across which they were characterized, and the classification each received from this study. The classification column is an outcome and is justified in §9. The range column is the characterization range, which is the edge of the knowledge space explored, and it is not a proven acceptable range. The proven acceptable ranges are computed per attribute in §7.

Table 4.1: Characterized parameters of the capture step, with set-points, normal operating ranges, characterization ranges and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	30	20–40	10–50	WC- CPP	multivari- ate
Elution buffer pH	pH	3.55	3.4–3.7	3.2–3.9	WC- CPP	multivari- ate
Load flow rate	cm/hr	200	150–250	100–300	KPP	multivari- ate
End of pool collect	CV	2.6	2.3–2.9	2–3.2	KPP	multivari- ate
Operating temperature	°C	22	18–26	15–30	GPP	univariate
Bed height	cm	20	15–25	10–30	GPP	univariate

The characterization ranges are two to three times the width of the routine control ranges. Protein load was studied from 10 to 50 g/L resin against a normal operating range of 20 to 40 g/L resin, and elution buffer pH from 3.2 to 3.9 pH against 3.4 to 3.7 pH. The lower edge of the elution pH range was set at 3.2, because platform experience indicates that antibody precipitation can occur below that value and a study that precipitated product would characterize a different step. The wider ranges give the process room to move inside a demonstrated region and support future changes within it.

The four multivariate parameters are coded A to D in the effect and coefficient tables throughout this report. Table 4.2 gives the assignment.

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

4.2 Screening design

The screening study was a two-level full factorial in the four multivariate parameters with 3 centre-point replicates, 19 runs in all, executed in randomized order. A full factorial was used rather than a fraction because four factors cost only 16 factorial runs, and because the interaction between protein load and elution pH was expected on mechanistic grounds and a fractional design would have aliased some two-factor interactions with each other. All main effects and all two-factor interactions are therefore estimated clear of one another. Interactions of three factors and above are not in the model and fall into the residual.

The screening model carries no quadratic term, so it identifies which parameters act and in which direction, and it does not describe curvature. The magnitudes it returns are refined by the response-surface model in §5.3, which is the predictive model of this report. The design matrix and the measured responses are in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same four parameters, 28 runs, made up of 16 factorial points, 8 axial points at the faces of the cube and 4 centre-point replicates. The axial points were placed on the faces rather than outside them, so that no run exceeds the characterization range of any parameter and every run remains a condition the step could be operated at. The design estimates all main effects, all two-factor interactions and all four pure quadratic terms.

All four screening factors were carried into the response-surface design. Two of them, load flow rate and end of pool collect, showed only small effects on the impurity responses in screening, and they were retained because both act on yield and because dropping a factor from the design would have held it at its set-point and hidden any curvature it carries. The design matrix and the measured responses are in Appendix B.

4.4 Univariate assessment

Operating temperature and bed height were assessed one at a time across 15 to 30 °C and 10 to 30 cm respectively, and neither was carried into the multivariate designs. Temperature acts on the diffusion coefficient of the antibody and on the strength of the hydrophobic contacts at the interface between the Fc region and the ligand, and both effects are small across the range studied compared with those of load and pH. The binding capacity of the resin is specified by its manufacturer across the whole range. Bed height at a fixed linear velocity sets the residence time and the pressure drop across the column, and in scale-up it is compensated by the linear velocity so that residence time is held. Within the range studied it is a property of the packing rather than of the chemistry.

Both parameters were held at their set-points during the multivariate runs. The univariate assessments therefore support the ranges of the two parameters themselves, and they do not test whether either interacts with protein load, elution buffer pH, load flow rate or end of pool collect. That limit is stated again in §11.

5 Results

5.1 Centre-point performance and reproducibility

The centre-point replicates were executed at the commercial set-point of all four multivariate parameters, so they measure the reproducibility of the scale-down model at the condition the process will be operated at. Table 5.1 gives the mean, the standard deviation and the coefficient of variation of each response over the 4 replicates of the response-surface design.

Table 5.1: Centre-point performance of the response-surface design, over 4 replicates at the set-point.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	7.56e+03	357	4.72
Step yield	4	0.946	0.0187	1.97
Leached Protein A (ppm)	4	2.86	0.967	33.8

Step yield is the most reproducible of the three responses, at 2.0 % of the mean. Pool host cell protein follows at 4.7 %, which is below the reported precision of the assay that measures it, so this estimate over 4 replicates does not resolve the run to run variability of the step itself. Leached Protein A is the least reproducible, at 33.8 % over a mean of 2.86 ppm. Its spread is consistent with the analytical contribution described in §3.3, and one of the replicate results was investigated and re-assayed under DEV-005-02 (§13.2).

The screening design carried 3 centre-point replicates. Their coefficients of variation were 19.9 % for pool host cell protein, 0.6 % for step yield and 15.5 % for leached Protein A. The host cell protein figure rests on 3 results and is a weak estimate of the pure error, which is one reason the response-surface design carried a fourth replicate. The pure error terms used in the lack of fit tests of §5.3 are the response-surface replicates.

5.2 Screening: factor effects

The screening models were fitted with all main effects and all two-factor interactions. Table 5.2 gives their adequacy.

Table 5.2: Screening model adequacy for each response.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.973	0.939	0.886	28.6	3.49e-05	1.4e+03
Step yield	19	0.976	0.945	0.798	32	2.28e-05	0.00833
Leached Protein A (ppm)	19	0.371	-0.414	-2.98	0.473	0.867	0.513

Two of the three responses are modelled well by the screening design and the third is not. Pool host cell protein and step yield give overall F tests significant at $p < 0.001$ and $p < 0.001$ respectively. The leached Protein A model returns an R^2 of 0.371 with an adjusted R^2 of -0.414 and an overall F test at $p = 0.867$, so no parameter acts on it detectably across the ranges studied. The negative adjusted R^2 means the ten estimated terms explain less than the mean of the data would.

Table 5.3 gives the six largest effects on pool host cell protein.

Table 5.3: Largest screening effects on pool host cell protein, in coded factors, where an effect is the change in the response across the full characterization range of the factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	

Pool host cell protein is dominated by two parameters. Raising elution buffer pH across its characterization range lowers pool host cell protein by 8,840 ng/mg ($p < 0.001$), and raising protein load across its range raises it by 7,269 ng/mg ($p < 0.001$). Both effects are an order of magnitude larger than the centre-point standard deviation of 357 ng/mg. The interaction between the two is significant ($p = 0.006$) and negative, which means the benefit of a higher elution pH is larger when the column is loaded closer to its capacity. Load flow rate and end of pool collect have no significant main effect on pool host cell protein in the screening design.

Table 5.4 gives the six largest effects on step yield.

Table 5.4: Largest screening effects on step yield, in coded factors.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*

Step yield is governed by the end of pool collection. Extending the collection across its characterization range raises yield by 6.0% ($p < 0.001$), which is more than three times the centre-point standard deviation of the response. Protein load and load flow rate both lower yield, by 2.6% and 2.4% across their ranges, and their interaction is significant ($p = 0.001$) and negative. A high load run at a high flow rate therefore loses more yield than the two main effects predict on their own. The two remaining significant terms, the interactions of load flow rate and of elution pH with the end of pool collection, are each at or below 1.1% across their ranges and are not carried into the interpretation.

The screening design identifies which parameters act and in which direction. It carries no quadratic term, so it cannot say whether a response turns within the range, and the effect sizes it returns are averages across the cube. The response-surface models in §5.3 refine both.

5.3 Response-surface models

Table 5.5 gives the adequacy of the full quadratic models fitted to the response-surface data.

Table 5.5: Response-surface model adequacy for each response.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.989	0.976	0.932	80.9	2.71e-10	895
Step yield	28	0.913	0.818	0.586	9.69	0.000105	0.016
Leached Protein A (ppm)	28	0.567	0.101	-0.714	1.22	0.365	0.637

The pool host cell protein model is the strongest of the three. It returns an R^2 of 0.989, an adjusted R^2 of 0.976 and a predicted R^2 from the PRESS statistic of 0.932. The gap between the adjusted and the predicted value is small, so the model is not over-fitted and predicts a withheld run about as well as it fits the data it was built on. The step yield model returns an R^2 of 0.913 with a predicted R^2 of 0.586. That gap is wider, and the yield model is used below for direction and magnitude and not for prediction at a point.

The leached Protein A model returns an overall F test at $p = 0.365$ and a predicted R^2 of -0.714. A negative predicted R^2 means the model predicts a withheld run worse than the mean of the data does. Leached Protein A is therefore reported as robust to

all four parameters across the ranges studied, and no response-surface model for it is used to make a claim in this report. The single term that reaches significance in it, the quadratic in elution pH, is discussed with the coefficient table in Appendix C.

Table 5.6 partitions the variation in the pool host cell protein model.

Table 5.6: Analysis of variance for the pool host cell protein response-surface model, with the residual split into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05	nan	nan
Lack of fit	1e+07	10	1e+06	7.88	0.058
Pure error	3.82e+05	3	1.27e+05	nan	nan

The model term is significant at $p < 0.001$. The lack of fit test compares the part of the residual the model cannot explain with the pure error of the centre-point replicates, and it returns $F = 7.88$ at $p = 0.058$. Lack of fit is therefore not significant at $\alpha = 0.05$, and the quadratic model is an adequate description of the surface. The test rests on 3 degrees of freedom of pure error, so it has limited power, and a lack of fit smaller than the one this design could detect would not have been found. The same test on the step yield model returns $p = 0.731$.

Table 5.7 gives the coefficients of the pool host cell protein model in coded factors.

Table 5.7: Coefficients of the full quadratic model for pool host cell protein, in coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7,215.5	309	23.3	5.44e-12	***
A	3,871.8	211	18.4	1.12e-10	***
B	-5,244.6	211	-24.9	2.4e-12	***
C	-435.4	211	-2.06	0.0596	
D	-1,055.5	211	-5	0.000242	***
AB	-1,899.8	224	-8.49	1.16e-06	***
AC	-429.1	224	-1.92	0.0774	
AD	-518.1	224	-2.32	0.0376	*
BC	348.2	224	1.56	0.144	
BD	727.4	224	3.25	0.00632	**
CD	-128.7	224	-0.575	0.575	
A ²	-344.5	557	-0.618	0.547	
B ²	1,943.5	557	3.49	0.00401	**
C ²	851.3	557	1.53	0.151	
D ²	330.5	557	0.593	0.563	

The two main effects confirm the screening result and sharpen it. Elution buffer pH carries a coefficient of -5,245 ng/mg per coded unit and protein load carries 3,872

ng/mg per coded unit, and both are significant at $p < 0.001$ and $p < 0.001$. Their interaction is significant at $p < 0.001$ with a coefficient of -1,900 ng/mg. The quadratic term in elution pH is positive and significant at $p = 0.004$, so the surface is curved in pH. Pool host cell protein falls steeply as pH rises from the low edge of the range and the fall flattens toward the high edge, which means the penalty for operating below the set-point is larger than the benefit of operating above it.

Two further terms reach significance and both are small. The end of pool collection lowers pool host cell protein by 1,055 ng/mg per coded unit ($p < 0.001$), and its interactions with protein load and with elution pH are significant at $p = 0.038$ and $p = 0.006$. Load flow rate does not reach significance in this model ($p = 0.060$).

Table 5.8 gives the coefficients of the step yield model.

Table 5.8: Coefficients of the full quadratic model for step yield, in coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.947	0.00554	171	3.5e-23	***
A	-0.0165	0.00377	-4.36	0.000767	***
B	-0.00243	0.00377	-0.643	0.531	
C	-0.00891	0.00377	-2.36	0.0346	*
D	0.0365	0.00377	9.68	2.62e-07	***
AB	0.00266	0.004	0.665	0.517	
AC	-0.00914	0.004	-2.28	0.0399	*
AD	0.00629	0.004	1.57	0.14	
BC	0.00243	0.004	0.606	0.555	
BD	0.00291	0.004	0.727	0.48	
CD	0.00219	0.004	0.547	0.594	
A ²	-0.0128	0.00997	-1.29	0.221	
B ²	0.012	0.00997	1.2	0.252	
C ²	0.00681	0.00997	0.683	0.507	
D ²	-0.0178	0.00997	-1.78	0.0976	

Step yield is governed by the end of pool collection, at 3.7% per coded unit ($p < 0.001$). Protein load costs 1.6% per coded unit ($p < 0.001$) and load flow rate costs 0.9% per coded unit ($p = 0.035$). Their interaction is significant at $p = 0.040$. No quadratic term reaches significance, so within the ranges studied yield does not turn.

The predicted responses at the set-point are 7,216 ng/mg for pool host cell protein, 94.7% for step yield and 2.87 ppm for leached Protein A. The host cell protein prediction sits a factor of 1.49 below the in-process limit of 10,724 ng/mg derived in §3.5.

Figure 5.1 shows the response surfaces over protein load and elution buffer pH with the other two parameters at their set-points, and Figure 5.2 shows them over load flow rate and end of pool collect. The reader should look at the first panel of Figure 5.1, where the contours run diagonally rather than parallel to either axis. That diagonal is the interaction, and it is the feature that constrains the design space in §6.

Response-surface predictions (remaining factors held at set-point)

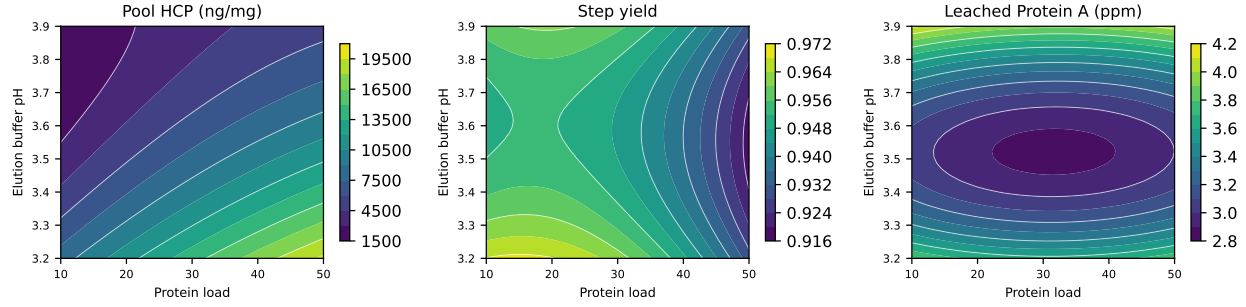


Figure 5.1: Response-surface predictions over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points.

Response-surface predictions (remaining factors held at set-point)

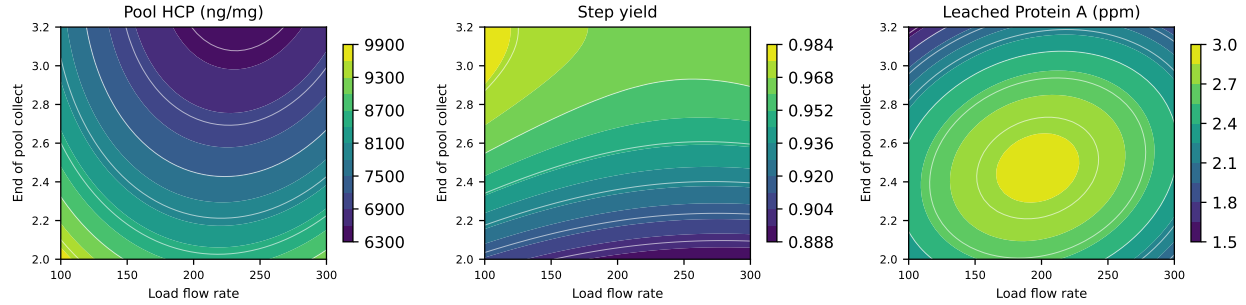


Figure 5.2: Response-surface predictions over load flow rate and end of pool collect, with protein load and elution buffer pH at their set-points.

Figure 5.3 gives the diagnostics of the pool host cell protein model, and Figure 5.4 those of the step yield model. Each shows the standardized residuals against the predicted values, the normal quantile plot of the residuals, and the actual values against the predicted ones. The residuals of the host cell protein model are spread evenly across the predicted range and the actual values lie close to the diagonal. The residuals of the yield model are larger in relative terms at the low end of the predicted range, which is where the design places the high-load and high-flow runs, and this is the behaviour that the wider gap between its adjusted and predicted R^2 reflects.

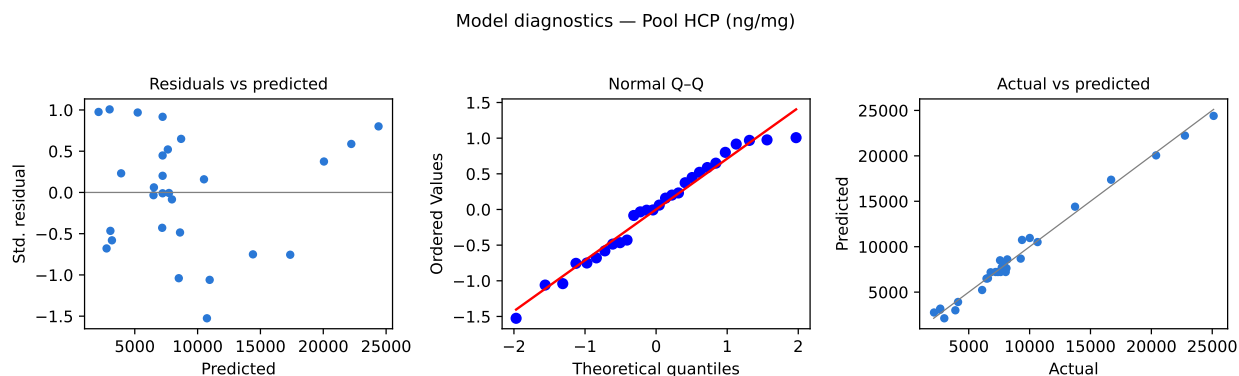


Figure 5.3: Model diagnostics for the pool host cell protein response-surface model.

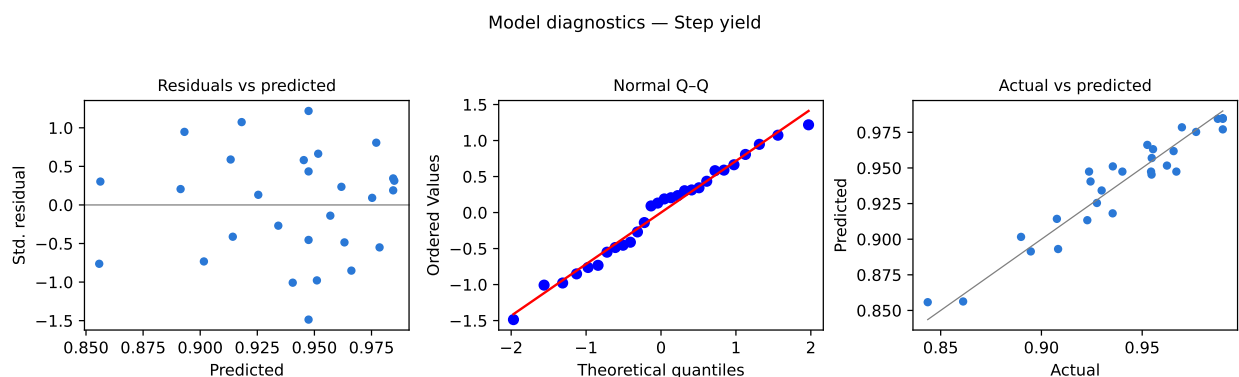


Figure 5.4: Model diagnostics for the step yield response-surface model.

5.4 Mechanistic interpretation

The surfaces follow from the physical chemistry of affinity capture. Host cell protein reaches the eluate by two routes. Some of it binds the resin backbone or the ligand weakly and elutes when the pH is lowered, and some associates with the antibody itself and is carried through as a complex. Both routes depend on how much antibody is bound and on how completely it is released. As the load approaches the dynamic binding capacity the mass transfer zone extends further down the bed, and impurity that the wash would otherwise have removed is carried into the eluate. A lower elution pH releases weakly bound host cell protein together with the product. The two effects therefore reinforce each other, and the interaction between load and elution pH that Table 5.7 reports is a property of the mechanism and not of the design.

The curvature in elution pH has the same origin. The histidine residues at the interface between the Fc region and the ligand are already largely protonated across the elution range, so within it a lower pH acts by driving the residual affinity of the complex down further and by protonating the carboxylate contacts that remain. The antibody then desorbs sooner and in a sharper peak, and the weakly held impurity is released

with it. Toward the high end of the range the peak is broader and later and carries less impurity, and the marginal benefit of a further increase falls away. That is the flattening the positive quadratic term describes.

The end of pool collection acts on the impurity ratio through the denominator. The weakly held impurities elute early in the low-pH transition and not in the tail, so extending the collection adds antibody mass without adding a proportionate impurity mass, and the concentration of host cell protein per unit of antibody falls. The same mechanism explains why extending the collection helps more at a high load, which is the negative interaction between load and end of pool collect, and less at a high elution pH, which is the positive interaction between pH and end of pool collect. Where there is little co-eluting impurity, there is little to dilute.

Yield behaves as the transport argument predicts. Antibody is lost when it breaks through during the load or is left on the column at the end of elution. Breakthrough rises as the load approaches the dynamic binding capacity and as residence time falls, because binding at the ligand is limited by diffusion into the pores of the bead. A high load run at a high flow rate therefore loses more than either condition costs alone, which is the negative interaction the yield model reports. Extending the pool collection recovers the tail of the elution peak and raises yield directly.

Leached Protein A behaved as the leaching mechanism predicts. Ligand leaves the resin by hydrolysis of the immobilization chemistry and by proteolysis, and the rate depends on the coupling chemistry, on the cumulative cycle number and on the sanitization history. None of those quantities is a parameter of a single run, and all were held constant across this study. The absence of a detectable parameter effect is therefore the expected result and not a failure of the design. It also means that leached Protein A is controlled by the resin life-cycle limits of SOP-2008 and by downstream clearance, and not by the operating parameters of this step.

6 Design space

The design space of the capture step is the region of the characterized ranges of the four multivariate parameters over which the two governed quality attributes meet the criteria that apply to them. Pool host cell protein must stay at or below the in-process limit of 10,724 ng/mg and leached Protein A at or below 5 ppm, both derived in §3.5. The region was evaluated by predicting both responses from the response-surface models of §5.3 on an evenly spaced grid of 14,641 points across the four characterization ranges.

73.4% of the characterized region meets both criteria. Pool host cell protein is the binding response, and it rejects 26.6% of the grid on its own. Leached Protein A rejects none of it, which follows from §5.3, where no parameter was found to act on that response and the predicted level stays far below its criterion everywhere in the region. The design space is therefore set by one response and by the two parameters that govern it.

The principal plane of the design space is protein load against elution buffer pH, and the constraint in that plane is joint rather than separate. With load flow rate and end of pool collect at their set-points, the highest protein load that meets the in-process limit is 17.7 g/L resin at the low edge of the elution pH range, 33.9 g/L resin at the low edge of the normal operating range, and 49.9 g/L resin at the set-point. At an elution pH of 3.7 and above, the whole characterized load range meets the limit. The acceptable region in this plane is a wedge that opens as elution pH rises, and its shape is the interaction between the two parameters reported in Table 5.7.

The normal operating ranges lie inside the design space over most, and not all, of their extent. Predicting pool host cell protein across the box formed by the four normal operating ranges gives values from 3,545 to 13,553 ng/mg, and 93.8% of that box meets the in-process limit. The part that does not is the corner at high protein load and low elution pH, and it is the same corner that the grid over the wider characterization ranges excludes. At the low edge of the elution pH range the mean model reaches the limit at a load of 33.9 g/L resin, which is below the upper edge of the load range of 40 g/L resin. Protein load and elution buffer pH are therefore not independent within their routine ranges, and the control strategy in §10 holds them as a pair rather than as two separate limits.

Two bounds apply to this design space. The predictions are mean levels, so the reliability of the process at the edge of the region is close to half if the variability of the response is symmetrical about the mean, and an assurance statement needs an interval. The proven acceptable ranges in §7 supply that interval and are correspondingly narrower. The second bound is scale. The region was defined on the qualified scale-down model, and its edges have not been confirmed at commercial scale.

Movement within the design space is not a change from a regulatory filing perspective (International Council for Harmonisation 2009). Planned movement is still assessed under the pharmaceutical quality system, with a prospective risk assessment of the particular move and a conclusion that existing knowledge supports it (International Council for Harmonisation 2008). Movement outside the region, including operation at a protein load above the wedge boundary at the elution pH in use, is a change and is handled under change control.

7 Proven acceptable ranges

A proven acceptable range is the range of one parameter over which the attribute in question meets its criterion when the other parameters are held or varied in a defined way. Two analyses were run for each governed attribute and each multivariate parameter, and Table 7.1 gives both. The first holds the other three parameters at their set-points and uses the mean prediction. The second lets them vary within their normal operating ranges and requires the 95 % predictive interval of the attribute to stay within the criterion, so it is a robustness statement and the narrower of the two.

Table 7.1: Proven acceptable ranges per governed quality attribute and parameter, at the set-point and propagated across the normal operating ranges.

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Protein load	10-50	g/L resin	$\leq$ 1.07e+04	10-49.8	10-34.5
Pool HCP (ng/mg)	Elution buffer pH	3.2-3.9	pH	$\leq$ 1.07e+04	3.36-3.9	3.48-3.9
Pool HCP (ng/mg)	Load flow rate	100-300	cm/hr	$\leq$ 1.07e+04	100-300	138-300
Pool HCP (ng/mg)	End of pool collect	2-3.2	CV	$\leq$ 1.07e+04	2-3.2	2.29-3.2
Leached Protein A (ppm)	Protein load	10-50	g/L resin	$\leq$ 5	10-50	10-50
Leached Protein A (ppm)	Elution buffer pH	3.2-3.9	pH	$\leq$ 5	3.2-3.9	3.2-3.86
Leached Protein A (ppm)	Load flow rate	100-300	cm/hr	$\leq$ 5	100-300	100-300
Leached Protein A (ppm)	End of pool collect	2-3.2	CV	$\leq$ 5	2-3.2	2-3.2

For pool host cell protein the analysis at the set-point returns almost the whole characterization range of protein load, 10 to 49.8 g/L resin, and a range for elution buffer pH of 3.36 to 3.9 that stops short of the low edge of the characterized range. Both results follow from the surface described in §5.3. With the other parameters at

their set-points there is enough headroom below the in-process limit for load to reach its upper edge, and the constraint appears only when elution pH is low.

The propagated analysis is stricter and it moves both ranges. The acceptable protein load becomes 10 to 34.5 g/L resin, which stops below the upper edge of the routine load range of 40 g/L resin. The acceptable elution buffer pH becomes 3.48 to 3.9, which stops above the lower edge of the routine pH range of 3.4. Two statements follow. Neither the top of the load range nor the bottom of the pH range is supported when the other parameters vary across their routine ranges and the predictive interval is required to stay within the limit. The pair is supported jointly, because the propagated analysis for each parameter varies the other one across its whole routine range, including the corner where the two act together. Load flow rate and end of pool collect are acceptable across 138 to 300 cm/hr and 2.29 to 3.2 CV under the same analysis, and both routine ranges lie inside those.

For leached Protein A every parameter is acceptable across its whole characterization range under the analysis at the set-point. Under the propagated analysis the elution buffer pH range becomes 3.2 to 3.86, and the remaining three parameters stay at their full ranges. That upper bound does not come from a demonstrated effect of pH, because §5.3 found none. It comes from the residual variability of a response whose model explains little, which the propagated analysis carries into the predictive interval, and from the fact that the drug substance criterion of 5 ppm is only a factor of 1.7 above the level the pool carries at the set-point. The mechanism in §5.4 places the control for leached Protein A on the resin life cycle, and that is where the control strategy puts it.

Figure 7.1 shows the analysis for pool host cell protein against elution buffer pH, its governing parameter, and Figure 7.2 shows it against protein load. Figure 7.3 shows leached Protein A against protein load. In each figure the left panel is the analysis at the set-point and the right panel is the propagated analysis, the acceptable region is shaded green, the acceptance limit is drawn as a dashed line, and the set-point and the normal operating range are marked. The right panel of Figure 7.2 is the one that carries the finding of this section, because the green region there ends inside the marked normal operating range.

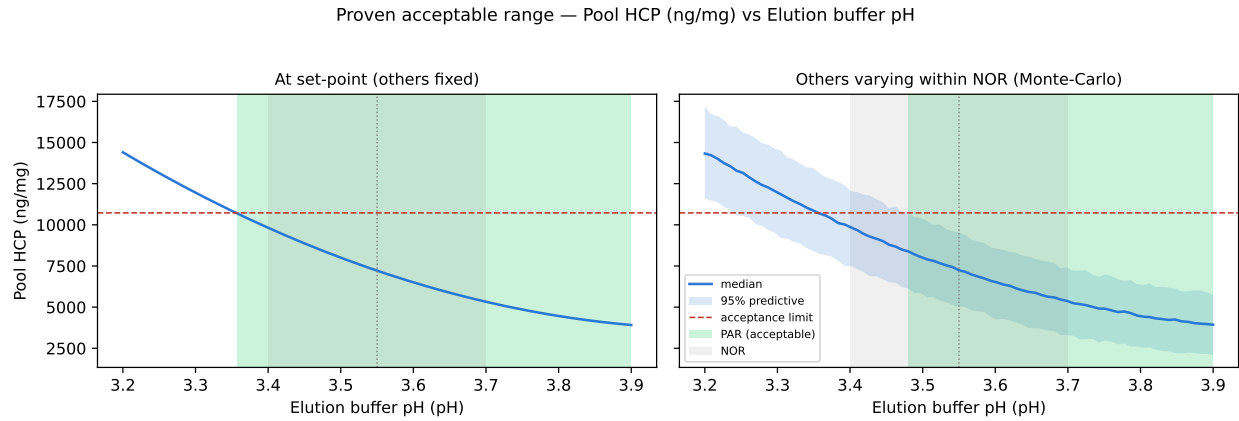


Figure 7.1: Proven acceptable range for pool host cell protein against elution buffer pH, at the set-point of the other parameters and propagated across their normal operating ranges.

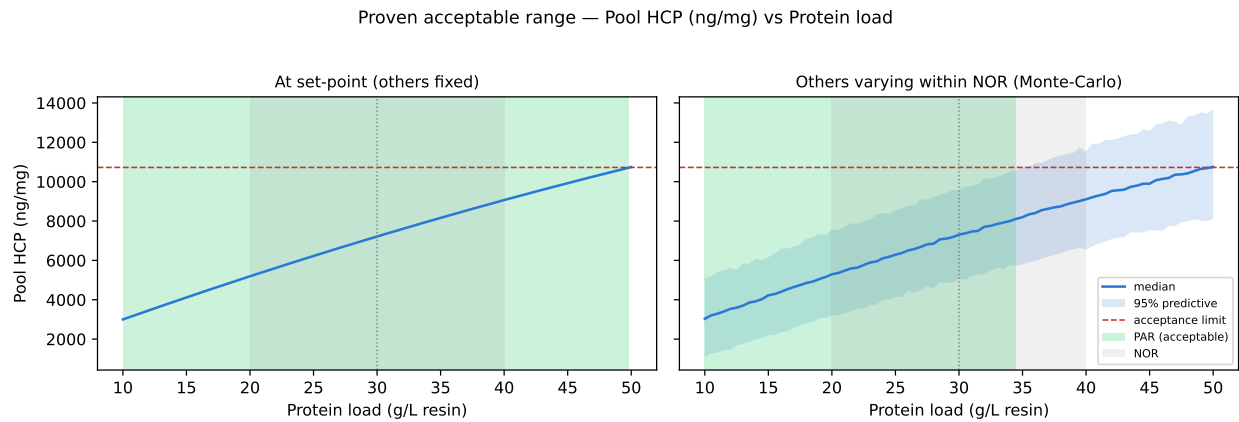


Figure 7.2: Proven acceptable range for pool host cell protein against protein load.

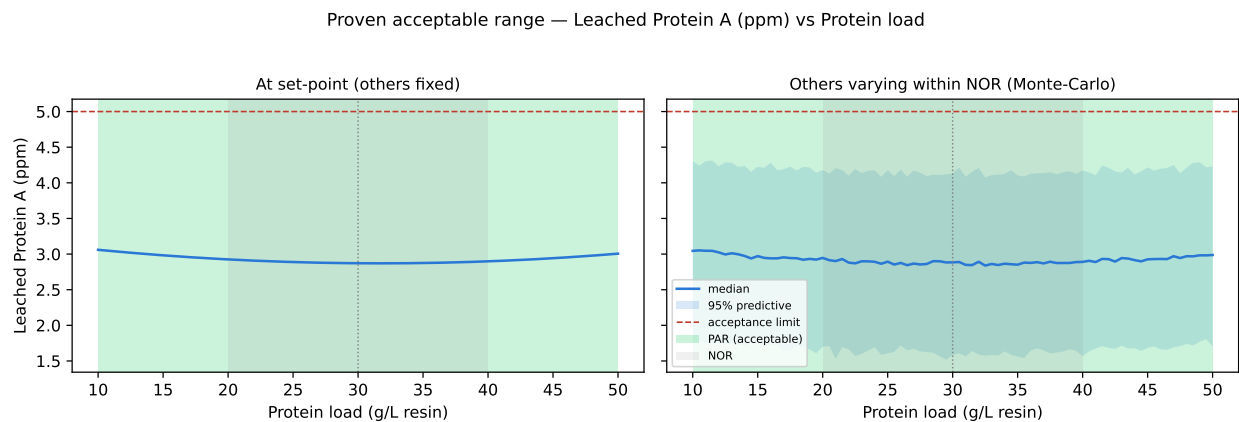


Figure 7.3: Proven acceptable range for leached Protein A against protein load, its governing parameter.

The proven acceptable ranges and the design space of §6 answer different questions and are used differently. A proven acceptable range is a statement about one parameter with the others defined, and it is the basis for assessing a single parameter excursion. The design space is the multivariate region, and it is what the process is operated inside. Where the two disagree, as they do for protein load, the narrower statement governs the control strategy.

8 Process capability and robustness

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with every parameter of the train drawn within its normal operating range and the fitted models of each step applied in series. Table 8.1 gives the result for the three impurity attributes this step governs.

Table 8.1: Commercial-scale capability for the impurity attributes governed by the capture step, over 2,000 simulated batches.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 $\pm$ 4.6	6.1
Residual DNA	VL	upper	0-0.001	0.0001 $\pm$ 0	10.8
Leached Protein A	L-M	upper	0-5	0.003 $\pm$ 0.0006	2,785.3

All three attributes meet their drug substance criteria across the simulated batches. Host cell protein is the tightest of the three, at a mean of 15.2 ng/mg against a criterion of 100 ng/mg and a one-sided Cpk of 6.14. That index is the one to read for this step, because host cell protein is also the response that binds the design space in §6 and the attribute whose in-process limit constrains the operating region.

The other two indices are large and are not informative as indices. Residual DNA reaches a Cpk of 10.8 and leached Protein A a Cpk of 2,785, because both attributes sit orders of magnitude below their criteria in the drug substance after the downstream clearance. The useful statement for leached Protein A is the margin rather than the index. The capture pool carries 2.87 ppm at the set-point, against a drug substance criterion of 5 ppm, and cation exchange and anion exchange reduce that further before the drug substance is formed.

The lowest capability index in the drug substance register is 1.51, and it belongs to the cumulative parvovirus clearance, which is set at anion exchange and reported in PCR-008. No capability of this step approaches it. No numerical acceptance threshold for Cpk is set in this package, and none is asserted here. The indices are reported with the mean and the criterion beside them, so the margin can be read directly.

Robustness is the other half of the picture, and the three responses differ. Leached Protein A is robust to all four multivariate parameters across their characterization ranges, in the strict sense that no term reached significance at $\alpha = 0.05$ in either design. Step yield varies from 90.6% to 97.2% across the normal operating range box, which is a performance range and not a quality risk. Pool host cell protein is not robust in that sense, and it is the attribute the control strategy manages.

9 Parameter classification

Parameters were classified under SOP-4001 from the effects demonstrated in this study and from the capability of the commercial equipment to hold each within the design space. A parameter whose variability affects a critical quality attribute is critical, and it is called well-controlled when the risk of falling outside the design space is low. A parameter that affects process consistency without affecting a quality attribute is a key process parameter. A parameter with no demonstrated effect on either is a general process parameter. The classification of each parameter is given in Table 4.1 and the reason for each follows.

- Protein load was classified as a well-controlled critical process parameter. It raises pool host cell protein by 7,269 ng/mg across its characterization range and it interacts with elution buffer pH, so its variability reaches a quality attribute. It is set by the measured titre of the clarified harvest and the loaded volume, both of which are measured before the load begins, so an excursion outside the design space is unlikely and detectable.
- Elution buffer pH was classified as a well-controlled critical process parameter. It carries the largest single effect on pool host cell protein in the study, 8,840 ng/mg across its characterization range. It is set by the composition of a buffer that is prepared and released before use under SOP-2103, and DEV-005-01 (§13.1) shows that a preparation error in that buffer is detected by the release test. The control is therefore effective and the risk of an undetected excursion is low.
- Load flow rate was classified as a key process parameter. It has no significant effect on pool host cell protein in the response-surface model ($p = 0.060$) and no significant effect on leached Protein A. It lowers step yield by 2.4% across its range and it interacts with protein load in doing so, so it is controlled for process consistency.
- End of pool collect was classified as a key process parameter. It carries the largest effect on step yield in the study, 6.0% across its characterization range. It also lowers pool host cell protein, by 1,055 ng/mg per coded unit, which acts in the direction of lower impurity and does not put the attribute at risk. Its whole characterization range meets the in-process limit under both proven acceptable range analyses (Table 7.1), so it is controlled for yield rather than for quality.
- Operating temperature was classified as a general process parameter. It was assessed one at a time across 15 to 30 °C, its mechanism of action on the affinity interaction is weak across that range, and the binding capacity of the resin is specified by the manufacturer across it.
- Bed height was classified as a general process parameter. It was assessed one at a time across 10 to 30 cm, and at a fixed linear velocity it sets residence time and pressure drop, both of which are held constant in scale-up by the packing specification of SOP-2008.

The capture step therefore has 2 well-controlled critical process parameters, 2 key process parameters and 2 general process parameters, and no critical process parameter. The absence of a critical process parameter reflects the control capability of the step and not the absence of an effect. Both well-controlled critical process parameters have large, demonstrated effects on pool host cell protein, and both are classified as well controlled because each is measured before the load begins and an excursion in either is detected. The joint constraint between them described in §6 does not change either classification, and it is carried into the control strategy in §10 as a paired limit rather than as two independent ranges.

10 Contribution to the control strategy

The capture step contributes to the control strategy through parameter ranges, through in-process monitoring of the pool, through the life-cycle control of the resin, and through the material it hands to the steps that follow. Each element rests on a result reported above.

Parameter ranges. The normal operating ranges in Table 4.1 are carried into Stage 2 for all six parameters. Protein load and elution buffer pH are controlled as a pair, because §6 shows that the acceptable load depends on the elution pH in use and §7 shows that neither routine range is supported across its full width when the other parameter varies. The paired limit is the wedge boundary of §6, and operation at a load above that boundary for the elution pH in use is a movement outside the design space.

In-process monitoring. Pool host cell protein is measured on every batch under AMV-3012 and is controlled against the in-process limit of 10,724 ng/mg derived in §3.5. That limit is not a specification and it is not the drug substance criterion. It is the level above which the downstream train would have to deliver more than its nominal clearance for the drug substance to meet its criterion, divided by an assurance margin of 4.5. Because it is derived from the clearance that cation exchange and anion exchange deliver, it moves if their performance changes, and it is re-derived whenever PCR-007 or PCR-008 is revised. Leached Protein A is measured on the pool under AMV-3016 and residual DNA under AMV-3014, both for monitoring.

Life-cycle control of the resin. §5.3 and §5.4 place the control for leached Protein A on the resin rather than on the operating parameters. The cycle limit, the sanitization sequence and the packing specification of SOP-2008 are therefore the controls that hold that attribute, and the study reported here does not substitute for them. This study was executed within those limits and does not characterize the effect of cumulative cycle number, which is established in the resin life-cycle study cited by SOP-2008.

Cross-step contributions. Host cell protein and leached Protein A are reduced further by cation exchange (PCR-007) and anion exchange (PCR-008), and the cumulative capability of the train for both is reported in Table 8.1. No viral clearance is claimed for this step, and the cumulative viral clearance claim rests on the low-pH hold (PCR-006), anion exchange (PCR-008) and small-virus retentive filtration (PCR-009). Residual DNA is removed here in the flow-through and the wash and further downstream, and no log reduction for it is claimed in this report.

Release testing. Host cell protein, residual DNA and leached Protein A are tested on the drug substance against the criteria in Table 2.1. The capability of the commercial process against those criteria is in Table 8.1. The control strategy for the whole train is consolidated in PCMR-001, and this report is one of its inputs.

11 Discussion

The capture step is now characterized to the depth the transfer requires. Two parameters govern the impurity burden of the eluate, protein load and elution buffer pH, and they act together rather than separately. The response-surface model for pool host cell protein describes that behaviour with an R^2 of 0.989, a predicted R^2 of 0.932 and no significant lack of fit, and it is the model the design space and the proven acceptable ranges rest on. Step yield is governed by the end of pool collection and is reduced by load and flow rate acting together. Leached Protein A is unaffected by all four parameters across the ranges studied, which moves its control to the resin life cycle.

The result that matters most for the control strategy is the one the two analyses of §7 disagree about. At the set-point of the other parameters, protein load is acceptable almost to the top of its characterization range. When the other parameters vary within their routine ranges and a predictive interval is required, it is acceptable only to 34.5 g/L resin, which is below the top of its routine range. The difference is not a discrepancy between two models. It is the difference between a mean prediction and an assurance statement, and the assurance statement is the one the control strategy uses.

Confidence in the scale-down model rests on its qualification against at-scale performance for the attributes of this step and on the reproducibility of its centre points, which is 4.7 % for pool host cell protein and 2.0 % for step yield. The model holds bed height, linear velocity, load per volume of resin and the buffer volumes in column volumes equal to the commercial values, so the chemistry at the two scales is the same. The claims made on it extend to the attributes it was qualified against and no further.

Two deviations were recorded and both were retained (§13). Neither changes a conclusion of this report. The buffer pH deviation affected one run of the response-surface design at a setting that carries no weight in the estimate of the elution pH main effect, and the response that run produced is consistent with its assigned setting. The leached Protein A deviation affected one result of a response that no parameter was found to act on.

Five limitations bound what this report claims.

1. The design space and the proven acceptable ranges were established on the scale-down model, and their edges have not been confirmed at commercial scale.
2. The design space of §6 is built from mean predictions. The assurance statement is in §7 and is narrower for protein load and for elution buffer pH.
3. The lack of fit test for the pool host cell protein model rests on 3 degrees of freedom of pure error, so a small lack of fit would not have been detected.

4. Operating temperature and bed height were assessed one at a time and were held at their set-points during the multivariate runs, so no interaction between either of them and the four multivariate parameters has been tested.
5. The in-process limit for pool host cell protein depends on the clearance that cation exchange and anion exchange deliver in the nominal train, so it is not a property of this step alone and is revised when theirs is.

The study covers the platform resin. RA-004 concluded that an alternate Protein A resin would require independent characterization with no bridging from platform data, and nothing in this report applies to that resin.

12 Conclusions

The Protein A capture step was characterized on a qualified scale-down model under PCP-005, with a 19 run screening design and a 28 run response-surface design in four parameters and two univariate assessments.

Pool host cell protein is governed by protein load and elution buffer pH acting together, and it is the response that binds the operating region. 73.4% of the characterized region meets the criteria for both governed attributes, and the excluded part is the corner at high load and low elution pH. Within the routine ranges the same corner is the only part that breaches the in-process limit, so protein load and elution buffer pH are carried into Stage 2 as a jointly controlled pair. Leached Protein A is unaffected by all four multivariate parameters across the ranges studied and is controlled by the resin life cycle. Step yield is governed by the end of pool collection.

2 parameters were classified as well-controlled critical process parameters, 2 as key process parameters and 2 as general process parameters. No parameter of this step was classified as a critical process parameter. At commercial scale all three impurity attributes this step governs meet their drug substance criteria, and the tightest of the three is host cell protein at a Cpk of 6.14. Two deviations were recorded and both were retained after investigation. The step is fit for transfer to Novacyte Biologics, Grafton, WI, and this report rolls up into PCMR-001.

13 Deviations from the plan

2 deviations from PCP-005 were recorded during execution. Both were investigated, both were dispositioned as retained, and neither invalidated a study or required a re-execution. Table 13.1 lists them.

Table 13.1: Deviations recorded during execution of PCP-005.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-005-01	Elution-buffer pH prepared below target on an RSM run	buffer release testing	retained
DEV-005-02	Leached Protein A result initially out-of-trend; re-assayed	analytical review	retained

13.1 DEV-005-01, elution buffer prepared below target

The elution buffer lot LOT-BUF-5290 was released at pH 3.38 against a target of 3.55, a shortfall of 0.17 pH units. In coded units the released value is -0.49, which is about half way to the low edge of the characterized range. The lot was used on run 23 of the response-surface design. The deviation was raised at buffer release testing, and the investigation attributed it to an error in determining the titration endpoint during preparation. Preparation and release of the buffer are governed by SOP-2103.

Two lines of evidence bound the impact. Run 23 is an axial point at the low level of end of pool collect, and its assigned level for elution buffer pH is the centre of the design. A run at the centre level of a factor contributes nothing to the least-squares estimate of that factor's main effect in this design, so the estimate of the elution pH coefficient in Table 5.7 does not depend on which of the two pH values that run received. Second, the pool host cell protein measured on the run is 8,167 ng/mg. The fitted model predicts 8,601 ng/mg at the run's assigned settings and 11,961 ng/mg if the elution pH is taken as the released value. The measured result sits within one root mean square error of 895 ng/mg of the first prediction and more than three root mean square errors below the second, so the response the column produced is consistent with the assigned setting and not with the released one.

The run was retained in the reported analysis. The investigation did not resolve whether the error lay in the preparation of the buffer or in the release measurement of it, and it did not need to, because both readings lead to the same disposition for the reasons above. The released value lies inside the characterized range for elution

buffer pH, so the run remains a condition the step could be operated at in any case. The control that detected the error, the release test on the buffer before use, is carried into the control strategy in §10.

13.2 DEV-005-02, leached Protein A result out of trend

A leached Protein A result on one eluate pool of the response-surface set was flagged as out of trend at analytical review. The sample was re-assayed under AMV-3016 and returned 1.6 ppm. The investigation attributed the first result to suspected carryover in the assay, and the re-assayed value was reported.

The impact on the study is bounded by what the response supports. Leached Protein A showed no significant effect of any parameter in either design (§5.2 and §5.3), and it is reported as robust to all four multivariate parameters rather than through a model, so a single reported value does not carry a coefficient or a design space boundary. The centre-point coefficient of variation for this response is 33.8 %, the largest of the three responses, and the method contributes a substantial share of it (§3.3). Neither the original nor the re-assayed result approaches the drug substance criterion of 5 ppm.

The deviation was dispositioned as retained. Its lasting contribution is to the analytical control of the step rather than to the process. The leached Protein A assay is the least precise of the methods applied at this step, and the pool result is monitored rather than controlled to a limit, for the reasons given in §10.

14 Appendices A-D

14.1 Appendix A — Screening design matrix and responses

Table 14.1 gives the coded settings of the 19 screening runs in the run order executed, and Table 14.2 gives the responses measured on each. The factor letters are those of Table 4.2.

Table 14.1: Screening design matrix in coded factors.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 14.2: Responses measured on the screening runs.

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	factorial	9,965.7	0.918	2.64
2	factorial	9,515.4	0.99	3
3	factorial	9,022.6	0.928	3.03
4	factorial	7,893.7	0.98	2.98
5	factorial	3,211.7	0.915	2.88

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
6	factorial	2,523.1	0.986	3.82
7	factorial	2,540.2	0.929	3.3
8	factorial	3,108.1	0.96	2.43
9	factorial	16,568.2	0.913	2.47
10	factorial	19,737.9	0.99	2.66
11	factorial	18,271.3	0.857	2.67
12	factorial	21,244.6	0.936	3.54
13	factorial	6,791.9	0.911	2.77
14	factorial	7,875.0	0.976	2.29
15	factorial	7,655.0	0.891	2.88
16	factorial	7,791.9	0.927	3.14
17	center	6,436.2	0.931	2.91
18	center	9,652.2	0.933	3.07
19	center	8,221.9	0.942	3.86

14.2 Appendix B — Response-surface design matrix and responses

Table 14.3 gives the coded settings of the 28 response-surface runs, and Table 14.4 gives the responses measured on each. The type column distinguishes the factorial points, the axial points at the faces of the cube and the centre-point replicates.

Table 14.3: Response-surface design matrix in coded factors.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0

Run	Type	A	B	C	D
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 14.4: Responses measured on the response-surface runs.

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	factorial	10,011.3	0.93	3.42
2	factorial	9,269.0	0.99	3.06
3	factorial	10,650.9	0.928	2.53
4	factorial	7,715.2	0.988	2.48
5	factorial	2,992.5	0.923	3.27
6	factorial	2,149.1	0.977	1.89
7	factorial	2,643.4	0.908	4.61
8	factorial	2,663.4	0.99	3.13
9	factorial	25,114.3	0.89	3.38
10	factorial	20,388.5	0.99	2.49
11	factorial	22,755.7	0.861	2.12
12	factorial	16,694.1	0.924	3.03
13	factorial	7,882.2	0.895	3.63
14	factorial	6,577.7	0.97	2.72
15	factorial	6,798.5	0.844	3.06
16	factorial	6,100.0	0.962	2.82
17	axial	3,900.5	0.935	2.71
18	axial	9,377.3	0.935	3.37
19	axial	13,731.8	0.966	3.77
20	axial	4,121.9	0.955	3.8
21	axial	7,571.5	0.955	2.3
22	axial	8,097.7	0.955	2.54
23	axial	8,167.0	0.908	2.07
24	axial	6,460.6	0.953	2.69
25	center	8,036.2	0.94	2.75
26	center	7,616.5	0.967	1.7
27	center	7,206.4	0.924	4.06
28	center	7,396.1	0.954	2.91

14.3 Appendix C — Full effect and coefficient tables

Table 14.5 to Table 14.7 give every term of the screening models, including the terms not significant at $\alpha = 0.05$ that the tables in §5.2 omit. Table 14.8 gives the full quadratic coefficients for leached Protein A, the response whose model is not used in §5.3.

Table 14.5: Screening effects on pool host cell protein, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	
BD	-433.1	-216.5	350	-0.619	0.553	
C	167.3	83.7	350	0.239	0.817	
CD	-70.6	-35.3	350	-0.101	0.922	
BC	6.1	3.0	350	0.00868	0.993	

Table 14.6: Screening effects on step yield, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*
AB	0.00443	0.00221	0.00208	1.06	0.319	
AD	0.0037	0.00185	0.00208	0.889	0.4	
BC	0.0035	0.00175	0.00208	0.84	0.425	
B	-0.00236	-0.00118	0.00208	-0.568	0.586	

Table 14.7: Screening effects on leached Protein A, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.329	0.165	0.128	1.28	0.235	
A	-0.208	-0.104	0.128	-0.811	0.441	
BD	-0.189	-0.0944	0.128	-0.736	0.482	
BC	-0.182	-0.0912	0.128	-0.712	0.497	
C	0.18	0.0898	0.128	0.701	0.503	
D	0.152	0.0758	0.128	0.591	0.571	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AB	-0.133	-0.0664	0.128	-0.518	0.618	
CD	-0.0987	-0.0494	0.128	-0.385	0.71	
B	0.0652	0.0326	0.128	0.254	0.806	
AD	0.0567	0.0284	0.128	0.221	0.83	

Table 14.8: Coefficients of the full quadratic model for leached Protein A, in coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.87	0.22	13	7.67e-09	***
A	-0.027	0.15	-0.18	0.86	
B	0.148	0.15	0.989	0.341	
C	0.00918	0.15	0.0612	0.952	
D	-0.21	0.15	-1.4	0.185	
AB	-0.0127	0.159	-0.0796	0.938	
AC	-0.143	0.159	-0.899	0.385	
AD	0.132	0.159	0.831	0.421	
BC	0.269	0.159	1.69	0.115	
BD	-0.226	0.159	-1.42	0.179	
CD	0.167	0.159	1.05	0.314	
A ²	0.162	0.397	0.408	0.69	
B ²	0.906	0.397	2.29	0.0397	*
C ²	-0.458	0.397	-1.15	0.269	
D ²	-0.502	0.397	-1.27	0.228	

One term in Table 14.8 reaches significance at $\alpha = 0.05$, the quadratic in elution buffer pH, with a coefficient of 0.91 ppm and $p = 0.040$. It is not carried into any conclusion. The model that contains it is not significant overall ($p = 0.365$), its predicted R^2 is -0.714, and one term in 15 reaching the $\alpha = 0.05$ threshold is what that threshold produces by chance. Leached Protein A is reported as robust to all four parameters, as stated in §5.3.

14.4 Appendix D — Analytical methods summary

Table 14.9 summarizes the validated methods applied to the eluate pool. Performance characteristics are given for the two methods that carry the responses of the designed experiments. The remaining two are applied for monitoring, and their performance characteristics are held in their own validation reports.

Table 14.9: Validated analytical methods applied at the capture step.

Method	Title	Precision (%)	Limit of quantitation	Accuracy (%)
AMV-3016	Leached Protein A by ELISA (ppm)	6.5	0.25 ppm	97.4
AMV-3012	Host-Cell Protein ELISA	9.5	1 ng/mg	95
AMV-3014	Residual DNA (qPCR)	—	—	—
AMV-3011	Size-Variants (SEC-HPLC)	—	—	—

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